kader Academy

Machine Safety Specialist (CE)

Tab 2: CE-process and documentation
Module 1: CE-conformity process

Tab2-1/5 20250401

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1

CE-conformity process

Integrate in project approach

To often the 'CE-route' is considered as a separate route

1. Machine is ready, let 'the CE-employee' do his MD round.
   - *CE employee encounters unacceptable risks or deviations from standard. What now?*
   - *An inherent measure is certainly not going te be considered anymore.*

2. The risk assessment is a document that must be made 'machine-specific' at the end.
   - *We have an older version of a similar machine. We make this specific to this machine so that it can be included in the TF.*
   - *What about "state of the art"? Do current standards impose stricter requirements? Do the limits of the machine of the past match those of the current machine?*

2

CE-conformity process

Integrate in project approach

To often the 'CE-route' is considered as a parallel route

3. There will be a fence around the machine so it will be all right.
   - *Oh, should that fence have been higher?*
   - *Nice that you have provided a light screen there, but it is really too close to the machine. The drive can't stop that quickly.*

4. The hardware engineering department is responsible for machine safety, they do the PL calculations.
   - *Why does this have to be solved with a safety function? Couldn't the force of the cylinder itself have been limited?*
   - *Which safety functions need to be realised and what is the PLr for these functions?*

CE-conformity process

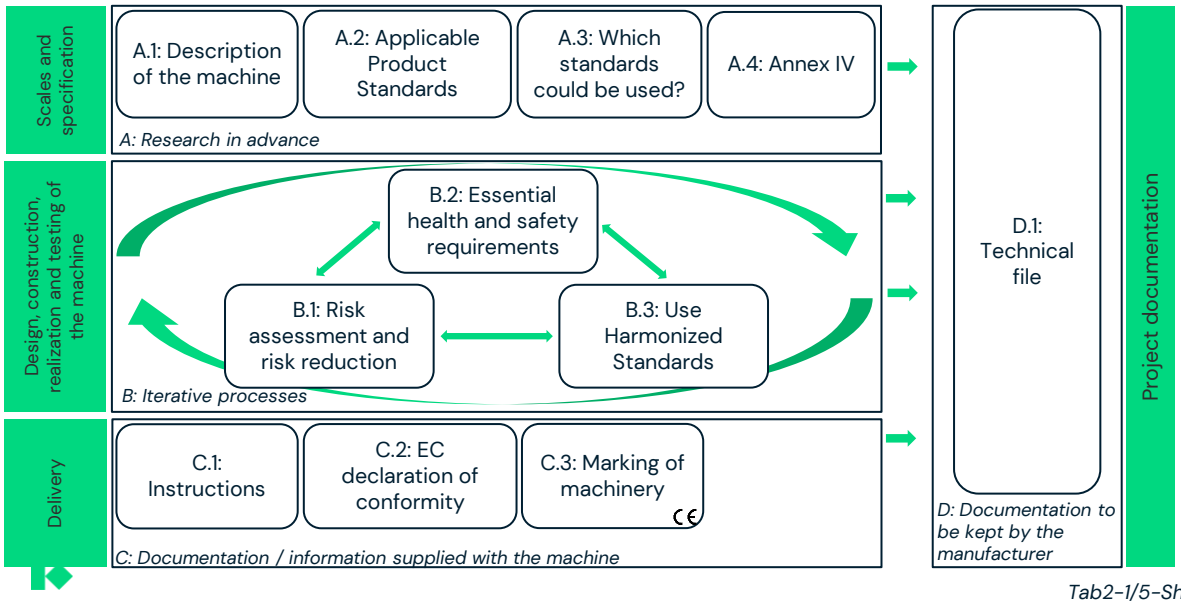
Integrate in project approach

In short:
Focus is on engineering, design and construction of the machine according to the client specification. CE process is too often considered as an afterthought or parallel documentation route.

Integrate the CE-process into the "normal" design process!
In all steps in the design and realization process of a machine, the relevant "CE-steps" must be integrated.

CE-conformity process

Integrate in project approach



Tab2-1/5-Sh.5

Tab2-1/5-Sh.6

CE-conformity process

Integrate in project approach

A.1: Description of the machine

- Define the operation and the (customer) specification of the machine (what should the machine do under which conditions).
- Consider the intended use and under what conditions the machine will be used.
- What is the expectation of the space in which the machine will be placed.
- What are the starting points for the product or material to be processed.
- Are there requirements for operating positions or manufacturing of components.
- Are there minimal requirements for 'safety aspects'.

Tab2-1/5-Sh.7

CE-conformity process

Integrate in project approach

Scales and specification

A.1: Description of the machine	A.2: Applicable Product Standards	A.3: Which standards could be used?	A.4: Annex IV

A: Research in advance

Example: description fictitious machine 'robot cell'

- The machine is a robot cell where products are supplied on a conveyor belt.
- At the end of the belt, the products are oriented and positioned by means of a cylinder.
- A robot picks up the product and places it into a production machine.

Tab2-1/5-Sh.8

CE-conformity process

Integrate in project approach

Scales and specification

A.1: Description of the machine	A.2: Applicable Product Standards	A.3: Which standards could be used?	A.4: Annex IV

A: Research in advance

Example: description fictitious machine 'robot cell'

Question: What aspects could / should be added to the very brief description on the previous sheet??.

Tab2-1/5-Sh.9

CE-conformity process

Integrate in project approach

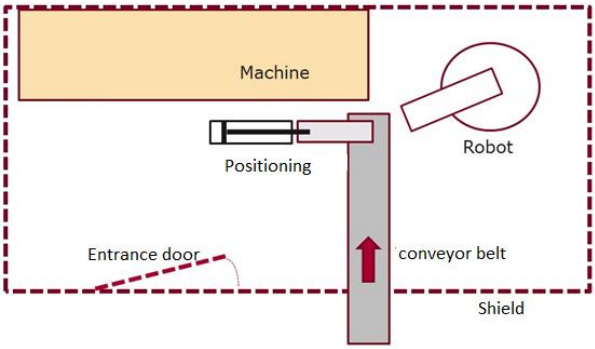
Scales and specification

A.1: Description of the machine	A.2: Applicable Product Standards	A.3: Which standards could be used?	A.4: Annex IV

A: Research in advance

Example: description fictitious machine 'robot cell'

- A layout or sketch of the machine can certainly be added value.



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Tab2-1/5-Sh.10

CE-conformity process

Integrate in project approach

Scales and specification

A.1: Description of the machine	A.2: Applicable Product Standards	A.3: Which standards could be used?	A.4: Annex IV

A: Research in advance

A.2: Applicable production directives

- For each machine, it will be necessary to determine which product directives are applicable at the front end of the process.
- The final machine will have to comply with all applicable requirements of all European Product Directives.

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CE-conformity process

Integrate in project approach

Scales and specification

A.1: Description of the machine	**A.2: Applicable Product Standards**	A.3: Which standards could be used?	A.4: Annex IV

A: Research in advance

Example: applicable production directives fictitious machine 'robotcell'

- 2006/42/EG Machine Directive → Yes
- 2014/30/EU EMC-Directive → Yes
- 2014/68/EU Pressure Equipment Directive (related to pneumatics)

→ most probably not due to category 'good craftmanship'.
Tip: Lloyd's Register Group has a handy app to determine this.
- 2014/34/EU Atex Directive → No
(the machine is not intended for use in an explosive atmosphere and does not create such an atmosphere itself)

CE-conformity process

Integrate in project approach

Scales and specification

A.1: Description of the machine	A.2: Applicable Product Standards	**A.3: Which standards could be used?**	A.4: Annex IV

A: Research in advance

A.3: Applicable standards

- It is useful to establish which standards 'must' be applied for the aspects that you already know will be applicable at the start of the project.
- Sometimes, the customer requirements already result in an overview of the required standards.
- The application of (harmonized) standards is usually the most effective way to sufficiently implement the essential health and safety requirements. If standards exist for a certain subject, but they are not applied, at least 'the same level of safety' must be guaranteed.

Tab2-1/5-Sh.13

CE-conformity process

Integrate in project approach

Scales and specification

A.1: Description of the machine	A.2: Applicable Product Standards	**A.3: Which standards could be used?**	A.4: Annex IV

A: Research in advance

Example: Applicable standards fictitious machine 'robot cell'

- EN 619: equipment for mechanical handling of unit loads [Type C standard]
- EN ISO 11161: Integrated manufacturing systems – Basic requirements[Type B standard]
- EN ISO 10218-2: Safety requirements for industrial robots - Part 2: Robot systems and Integration [Type C standard]
- EN ISO 14120: Guards - General requirements for the design and construction of fixed and movable guards[Type B standard]
- EN ISO 13857: Safety distances to prevent hazard zones being reached by upper and lower limbs [Type B standard]
- EN IEC 60204-1: Electrical equipment of machines - Part 1: General requirements [Type B standard]
- EN ISO 4414: Pneumatic fluid power - General rules and safety requirements for systems and their components [Type B standard]

Tab2-1/5-Sh.14

CE-conformity process

Integrate in project approach

Scales and specification

A.1: Description of the machine	A.2: Applicable Product Standards	**A.3: Which standards could be used?**	A.4: Annex IV

A: Research in advance

Example: Applicable standards fictitious machine 'robot cell'

- .....
- .....
- .....
- .....

Can you name some more standards (or subjects for which there is a standard) that might be applicable?

Tab2-1/5-Sh.15

CE-conformity process

Integrate in project approach

Scales and specification

A.1: Description of the machine | A.2: Applicable Product Standards | A.3: Which standards could be used? | **A.4: Annex IV**

A: Research in advance

A.4: Is the machine referred to in Annex IV of the MD?

Mentioned in Annex IV of the Machine Directive, for example:

- Portable chainsaws for woodworking.
- Woodworking machines.
- Presses; stroke > 6 mm and speed > 30 mm/s.
- Injection or compression plastics-moulding machinery.
- Machinery for underground working.
- Manually loaded trucks for the collection of household.
- Removable mechanical transmission devices.
- Vehicle services lifts.
- Devices for the lifting of persons or of persons and goods involving a hazard of falling from a vertical height of more than three metres.
- Protective devices designed to detect the presence of persons.
- Logic units for safety functions.
- Roll-over protective structures (ROPS).
- Falling-object protective structures (FOPS).

Tab2-1/5-Sh.16

CE-conformity process

Integrate in project approach

Scales and specification

A.1: Description of the machine | A.2: Applicable Product Standards | A.3: Which standards could be used? | **A.4: Annex IV**

A: Research in advance

A.4: Is the machine referred to in Annex IV of the MD?

- If the machine is not listed in Annex IV of the Machine Directive and no other product directives are applicable [A.2] that require a NoBo, then the machine may always be CE marked by the manufacturer himself, without involving a NoBo.
- If the machine is referred to in Annex IV of the Machine Directive, the machine must be manufactured in conformity with the applicable type C standards in order to be able to CE mark the machine by the manufacturer himself without having to involve a NoBo.
- If the machine is referred to in Annex IV of the Machine Directive and there is no type-C standard or one is not applied, then a NoBo has to be involved in order to be able to CE mark the machine.

CE-conformity process

Integrate in project approach

Scales and specification

A.1: Description of the machine | A.2: Applicable Product Standards | A.3: Which standards could be used? | A.4: Annex IV

A: Research in advance

Assessment procedures

Decision No:768/2008/EC

- Inspection modules, depending on the risk
- Covering all product directives
- Government-appointed (accredited) inspection bodies

Specific for Machine Directive

- About 95 % Inspection module A (do it yourself)
- **Annex IV machines**
  - option 1 C-standard applied→ Inspection module A
  - option 2 no C-standard applied → Inspection module B or H (NoBo)

Inspection Modules

	Code	Description	Notified Body	Qualitysystem
MD	A	Internal manufacturing control	N/A	
	A bis	Internal manufacturing control	specific test	
MD	B	EC type-examination	assessment and declaration of EC type-examination	
	C	Conformity to type	specific tests	
	D	Production quality assurance	inspection and surveillance of quality system	EN 29002 (ISO 9001:2000)
	E	Production quality assurance	inspection and surveillance of quality system	EN 29003 (ISO 9001:2000)
	F	Product verification	verification and certificate of conformity	
	G	EG-Unit verification	verification and certificate of conformity	
MD	H	Full quality assurance	surveillance of quality system, design examination, EC design examination certificate	EN 29001 (ISO 9001:2000)

CE-conformity process

Integrate in project approach

Notified Body – NoBo

Machine Directive (2006/42/EC), Article 14

- Assessment of conformity with requirements Directive
- Notified Body for category of machinery
- Well trained, equipped with technical means
- Identification number
- Assurance of independence
- Insured against civil liability
- Professional secrecy
- Verification of compliance with requirements (Annex IX (2006/42/EC))
- Publication in the official Journal of the European Union
- Authority to withdraw or suspend the Declaration of Conformity
- Exchange of experience

http://ec.europa.eu/growth/tools-databases/nando/

Tab2-1/5-Sh.19

Tab2-1/5-Sh.20

CE-conformity process

Integrate in project approach

Statement 1

After inspection by a Notified Body of a machine produced by me, my liability shall cease.

True / False ?

Tab2-1/5-Sh.21

CE-conformity process

Integrate in project approach

Scales and specification	A.1: Description of the machine	A.2: Applicable Product Standards	A.3: Which standards could be used?	A.4: Annex IV
	A: Research in advance			

Example: Applicable standards fictitious machine 'robot cell' in Annex IV?

Does the fictitious machine 'robot cell' fall under Annex IV?

Tab2-1/5-Sh.22

CE-conformity process

Integrate in project approach

Purchase and sale agreements
When assembly machinery or using completed or partly completed machinery in the new machine or assembly, it may be necessary to consider this in time:

- Supplier sometimes also delivers with CE-marking
- Chain liability
- TF of a subproduct can remain with supplier

Suggestions for inclusion in contract:

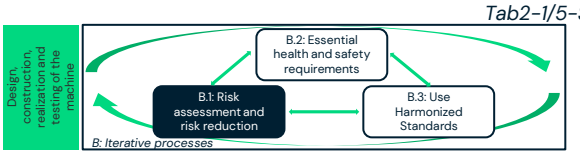
- Involve testing authority yes or no
- Complete TF and retain it for 10 years
- Which directives apply
- Which standards
- Methodology risk assessment
- Which test and trials and certificates
- User manual requirements
- User manual language

Requirements are especially important if you are combining subprojects into a complete production line or even plant

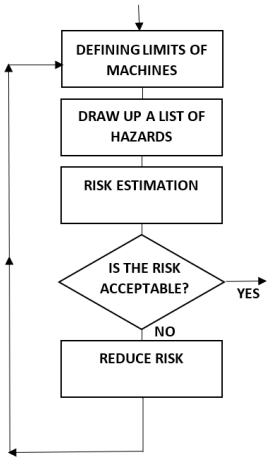
Tab2-1/5-Sh.23

CE-conformity process

Integrate in project approach



B.1: Risk assessment and risk reduction (see Tab 3 – module 1)

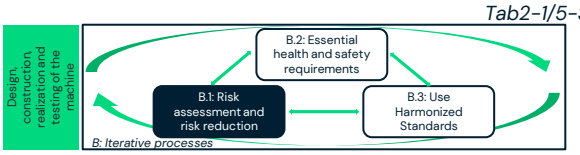


- On the left is a simplified representation of the risk assessment and risk reduction process.
- When the design of the machine is started, the risk assessment process should also be started.
- Make sure that this process is as integral as possible to the design of the machine.
- Regularly discuss with the multidisciplinary team whether the risk assessment is still up to date or whether adjustments are necessary.
- The risk assessment process can only be completed once the machine is ready.
- Think about version management and recording

Tab2-1/5-Sh.24

CE-conformity process

Integrate in project approach



B.1: Risk assessment and risk reduction

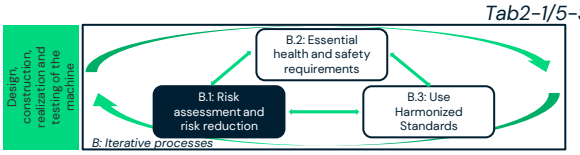
Risk-estimation (see Tab 3 – module 2)

- Risk graph, where the arguments weigh differently
- Lineair method, in which all arguments have equal weight
- Hybrid method

Tab2-1/5-Sh.25

CE-conformity process

Integrate in project approach



B.1: Risk assessment and risk reduction

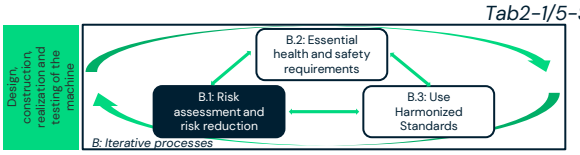
Risk-evaluation and risk reduction

- Basic principle: Eliminate hazards as far as possible or reduce risks as far as possible.
- Manufacturer is responsible for determining whether a residual risk is acceptable.
- For this purpose, it is very useful to use standards as they define the 'accepted' minimum safety level [B.3].
- In the risk assessment, state per residual risk why this is acceptable (otherwise further risk reduction is required).
- Define measures per risk in accordance with the 'imperative sequence'.
  - What was that order again?

Tab2-1/5-Sh.26

CE-conformity process

Integrate in project approach



B.1: Risk assessment and risk reduction

Risk-evaluation and risk reduction

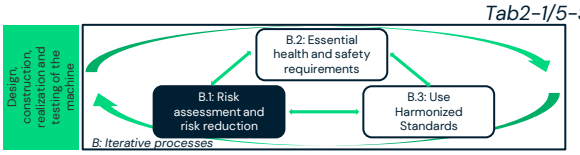
Mandatory sequence (Three step method):

1. Inherent measure (eliminate hazards in design)
2. Protection or shielding (reduce probability and exposure)
3. Information to the user (warning, acoustic signal, manual)

Tab2-1/5-Sh.27

CE-conformity process

Integrate in project approach



B.1: Risk assessment and risk reduction

Risk-evaluation and risk reduction

- By integrating this process of risk assessment and risk reduction into the design of the machine, the higher the chance of a good inherent measure.
- For each risk, describe the measures to be taken. It is essential to ensure that these measures are actually included in the relevant designs.
- It is therefore advisable to also uniquely code the measures so that during / after the design phase it can be verified that all measures are correctly included in the relevant (sub)designs.

Tab2-1/5-Sh.28

CE-conformity process

Integrate in project approach

B.1: Risk assessment and risk reduction

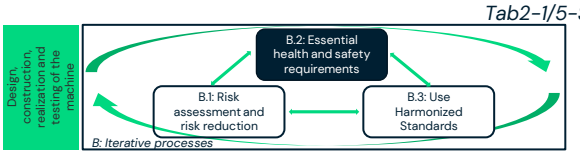
Example: Risk reduction fictitious machine 'robotcell

Explanation risk assessment	S	F	W	A	Risk	Protective measures			
Summary of the scenario factors (hazard analysis) with an explanation of the risk assessment.	S1 S2 S3	F1 F2	P1 P2 P3	A1 A2	1 - 14	STEP 1 Inherently safe design measures	STAP 2a Safeguarding and/or complementary protective measures	STAP 2b Identification safety functions	STAP 3 Information for use
S2: Broken bones in hands and/or fingers. F2: Constantly operational W2: Possible that an operator reaches through the guard in an unguarded moment. P2: Movement of robot is fast. Direction of movement not always known by the operator.	S2	F2	W2	P2	8		1-2-A Make the passage in the guard as small / low as possible that the largest product can just pass through. 1-2-B: Design the machine according to EN-ISO 13857, so that the distance from the opening in the guard to the maximum reach of the robot gripper is at least 180mm. 1-2-C: Provide an opportunity for the operator to stop the conveyor and control it in the other direction.		1-3-A: Warn in the instructions that the operators should not try to reach through the opening.

Tab2-1/5-Sh.29

CE-conformity process

Integrate in project approach



B.2: Essential health and safety requirements

- As with the risk assessment, it is important to start at the initial design stage by identifying which essential requirements, are applicable and which design discipline, if any, is concerned.
- Important link between risk assessment and essential requirements:

ANNEX I

Essential health and safety requirements relating to the the design and construction of machinery

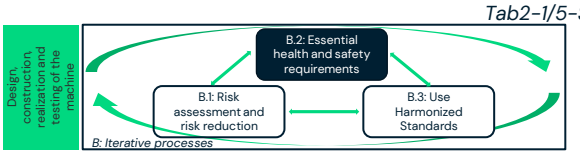
GENERAL PRINCIPLES

1. The manufacturer of machinery or his authorised representative must ensure that a risk assessment is carried out in order to determine the health and safety requirements which apply to the machinery. The machinery must then be designed and constructed taking into account the results of the risk assessment.

Tab2-1/5-Sh.30

CE-conformity process

Integrate in project approach



B.2: Essential health and safety requirements

- Example:

1.3.7. ***Risks related to moving parts***

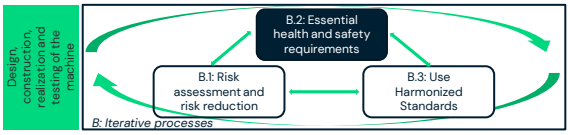
The moving parts of machinery must be designed and constructed in such a way as to prevent risks of contact which could lead to accidents or must, where risks persist, be fitted with guards or protective devices.

- This means that not all moving parts in a machine must necessarily be guarded (look at a conveyor belt). The point is that there must be no (unacceptable) risks related to moving parts in the machine.
- Use a customised template for the Verification & Validation of the essential requirements where a link to the risk assessment can also be made

Tab2-1/5-Sh.31

CE-conformity process

Integrate in project approach



B.2: Essential health and safety requirements

Example of Left part of template:

Requirement ID [2006/42/EG Annex I §.x]	Requirement text	Allocation of requirements to partial designs					
1		Requirement not applicable	Mechanical	Electrical	Software	n.a.	n.a.
1.3.7.	**Bewegende delen**						
1.3.7.a	The moving parts of machinery must be designed and constructed in such a way						
		x					

Here it can be checked if a requirement is not applicable to the machine..

Different parties or disciplines can be defined (Mechanical, Electrical, Software, etc.)

Tab2-1/5-Sh.32

CE-conformity process

Integrate in project approach

B.2: Essential health and safety requirements

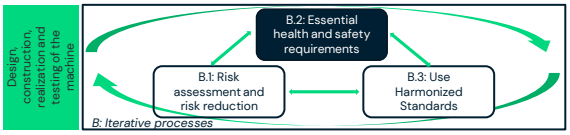
Example of right part of the template:

Requirement included / demonstrated in design [Y/N]?					Reference to relevant documents TF:	Applied (harmonised) standards:	Ref. to risk	Machine in accordance with requirement [Y/N]?		Substantiation / explanation / comment / action:
Mechanical	Electrical	Software	n.a.	n.a		No standards are mentioned in this column for requirements that refer to risks. These standards are already referred to in the risk assessment for the relevant risks.				In general, the risk assessment (at the referenced risks under the columns "protective measures" and "comment") indicates what still needs to be done to meet an essential requirement and to what extent this can beproved in the present situation.

Tab2-1/5-Sh.33

CE-conformity process

Integrate in project approach



Example: essential requirements fictitious machine 'robot cell'

V&V report Essential health and safety requirements Machinery Directive 2006/42/EG								Design verification - [Initials,date]					
								[Initials,date]	[Initials,date]	[Initials,date]	[Initials,date]	[Initials,date]	
Requirement ID [2006/42/EG Annex I §.x]	Requirement text	Allocation of requirements to partial designs						Requirement included / demonstrated in design [Y/N]?					Reference to relevant documents TF:
1	COMMON PRINCIPLES	Requirement not applicable	Mechanical	Electrical	Software	n.a.	n.a.	Mechanical	Electrical	Software	n.a.	n.a.	
1.3.7.	Bewegende delen												
1.3.7.a	The moving parts of machinery must be designed and constructed in such a way												
	- as to prevent risks of contact which could lead to accidents,	x											
	- or must, where risks persist, be fitted with guards or protective devices.		x	x	x			x	x	x			1. WTB design 2. Design SF 3. Electrical drawings 4. Software design 5. (i)FAT / (i)SAT / SIT reports

Tab2-1/5-Sh.34

CE-conformity process

Integrate in project approach

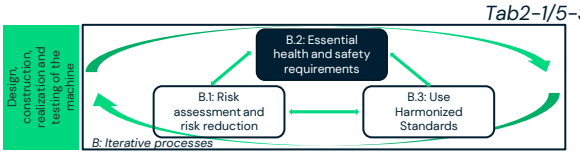
Example: essential requirements fictitious machine 'robotcell'

		Validation - [Initials,date]		
		[Initials,date]		
Applied (harmonised) standards:	Ref. to risk	Machine in accordance with requirement [Y/N]?		Substantiation / explanation / comment / action:
No standards are mentioned in this column for requirements that refer to risks. These standards are already referred to in the risk assessment for the relevant risks.				In general, the risk assessment (at the referenced risks under the columns "protective measures" and "comment") indicates what still needs to be done to meet an essential requirement and to what extent this can beproved in the present situation.
	1, 3, 25, 29	x		

Tab2-1/5-Sh.35

CE-conformity process

Integrate in project approach



Machine Directive Annex I

Essential health and safety requirements

- General essential health and safety requirements
- Supplementary essential requirements:
  - For certain categories of machinery
  - Mobility of machinery
  - Lifting operations
  - Machinery for underground working
  - Lifting or moving persons

Tab2-1/5-Sh.36

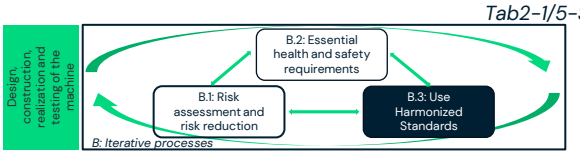
Exercise 1

- Check the Essential Requirements of the Machine Directive
- Determine which requirements contain concrete information and which requirements are (too) vague to determine compliance.
- Determine for the case you brought which requirements are applicable in any case.
- Plenary discussion

Tab2-1/5-Sh.37

CE-conformity process

Integrate in project approach



B.3: Harmonised standards *(See Tab 4)*

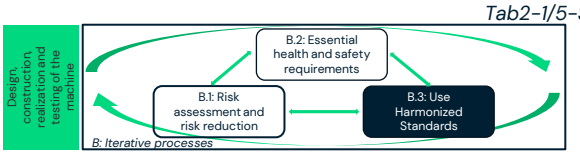
Apply harmonised standards where possible in order to both:

- related to the risk reduction measures [B.1], to be able to apply measures which are considered 'generally accepted' / 'good enough' and;
- achieve a presumption of conformity with the relevant applicable essential requirements [B.2].

Tab2-1/5-Sh.38

CE-conformity process

Integrate in project approach



Testing and trials
Examples
Machine Directive:

- Noise when > 70 dB(A)
- Safety functions
- Stability
- Emissions of hazardous substances
- Functional testing
- Lifting machinery: static and dynamic tests

Low voltage directive

- Earthing test
- Residual Current Protection Test

Pressure Equipment Directive:

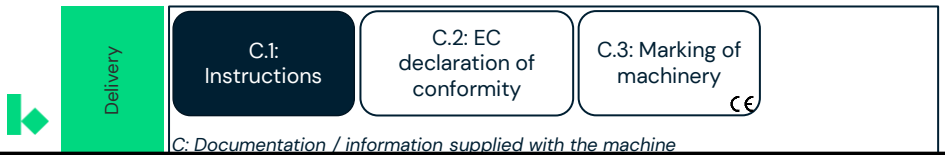
- Commissioning inspection of pressure equipment (depending on the category of pressure equipment)

CE-conformity process

Integrate in project approach

C.1: Instruction manual (See Tab2 module 2)

- All information required to use the machine safely throughout its lifetime and in all phases of its use.
  - for example, transport data, instructions for operators, diagrams, drawings, maintenance data, etc.
- Start drafting the instructions in time so that they are ready when the machine is delivered.
- Check whether all information required by the applied harmonised standards is included.

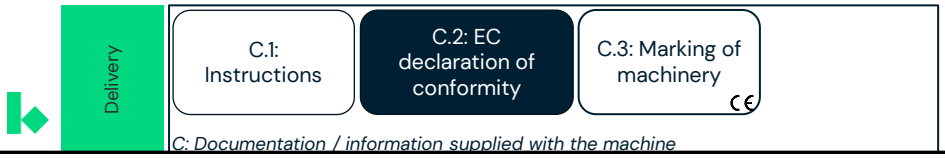


CE-conformity process

Integrate in project approach

C.2: EC-declaration of conformity (See Tab2 module 4)

- This is the important legal document in which the manufacturer declares that the machine complies with the applicable product directives [A.2] and which standards have been applied [A.3 & B.3].
- This declaration is formally signed by the manufacturer (usually the manager) and must be provided to the customer.

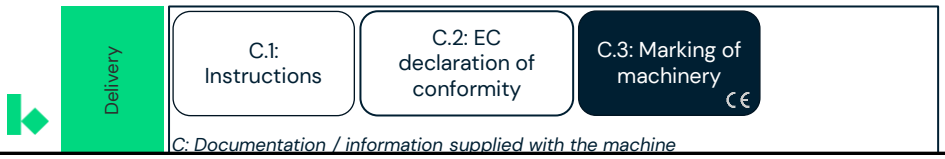


CE-conformity process

Integrate in project approach

C.3: Type plate and CE-marking (See Tab2 module 5)

- Finally, the machinery must bear the required CE marking.
- This must be affixed near the name of the manufacturer using the same technique (if the manufacturer is engraved, the CE marking must also be engraved).
- If a Notified Body is involved [A.4], this identification number will also have to be indicated. Other relevant aspects such as year of manufacture, consumption, possibly mass, should also be indicated here.

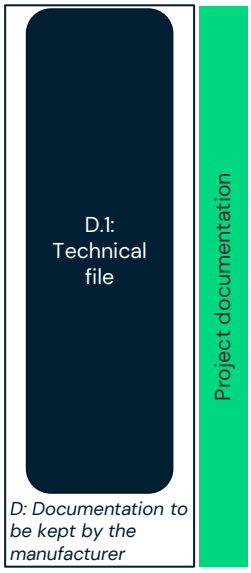


CE-conformity process

Integrate in project approach

D.1: Technical file (See Tab2 module 3)

- This is the collection of documentation recorded throughout the process.
- This file must demonstrate that the CE marking has been affixed to the machine correctly.
- It must therefore contain evidence that the applicable essential requirements have actually been met to a sufficient extent.
- The Technical File must be kept for at least ten years and in case of series production for ten years after the production of the last machine.
- Do not go through each document to determine whether it is really necessary to put it in the TF but include all documents and any relevant emails relating to the design and realisation process of the machine.



CE-conformity process

Integrate in project approach

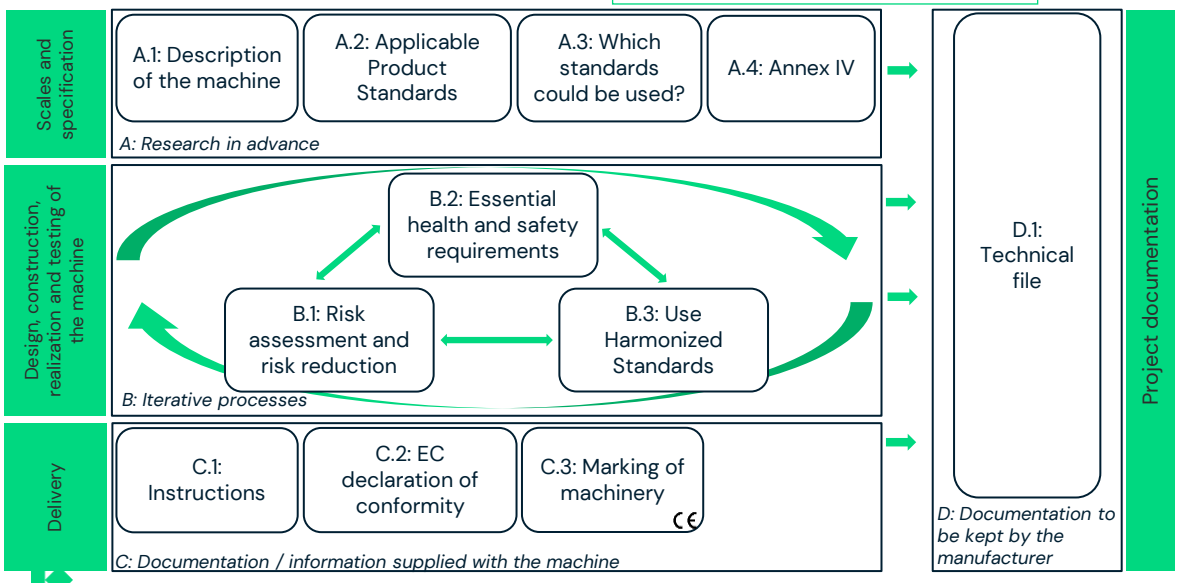
Documentation (partly completed) machinery

- For machinery:
  - Instruction manual (supplied with the machine)
  - Technical file
  - Declaration of Conformity (supplied with the machine)
  - Information on the machinery
- For partly completed machinery
  - Assembly manual (supplied with the partly completed machinery)
  - Relevant technical documents
  - Declaration of Incorporation (supplied with the partly completed machinery)
  - Information on the machinery

CE-conformity process

Integrate in project approach

It is important to coordinate and integrate this process into the 'normal engineering process'!



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Tab2-1/5-Sh.45

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